

Worked example: Ads × RPG interaction

Outcome 1 — Install success | Pooled analysis

1 Construct the two exposure-by-outcome tables

RPG stratum (n = 133,476)

	Ads = 1	Ads = 0	Total
Success	20,324	51,271	71,595
Not success	39,384	22,497	61,881
Total	59,708	73,768	133,476

Non-RPG stratum (n = 1,645,224)

	Ads = 1	Ads = 0	Total
Success	553,657	264,402	818,059
Not success	493,329	333,836	827,165
Total	1,046,986	598,238	1,645,224

2 Estimate conditional success probabilities

RPG

Outcome	Ads = 1	Ads = 0
Success	$20,324 / 59,708 = 0.3404$	$51,271 / 73,768 = 0.6950$

Non-RPG

Outcome	Ads = 1	Ads = 0
Success	$553,657 / 1,046,986 = 0.5288$	$264,402 / 598,238 = 0.4420$

$$\Delta p_{\text{RPG}} = 0.3404 - 0.6950 = -0.3546$$

$$OR_{\text{RPG}} = \frac{20324 \times 22497}{39384 \times 51271} = 0.2264$$

$$\Delta p_{\text{Non-RPG}} = 0.5288 - 0.4420 = +0.0868$$

$$OR_{\text{Non-RPG}} = \frac{553657 \times 333836}{493329 \times 264402} = 1.4170$$

Pooled interaction analysis

Install success | Additive and multiplicative scales

3 Additive interaction

$$DDP = \Delta p_{RPG} - \Delta p_{Non-RPG}$$

$$= -0.3546 - (+0.0868) = \mathbf{-0.4415}$$

95% CI: [-0.4468, -0.4362]

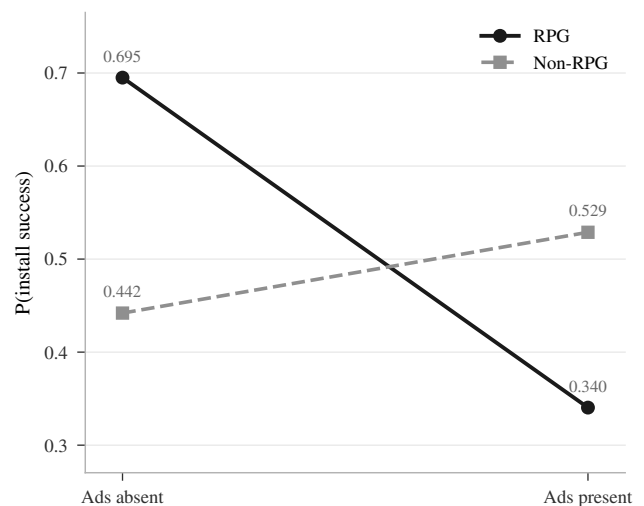
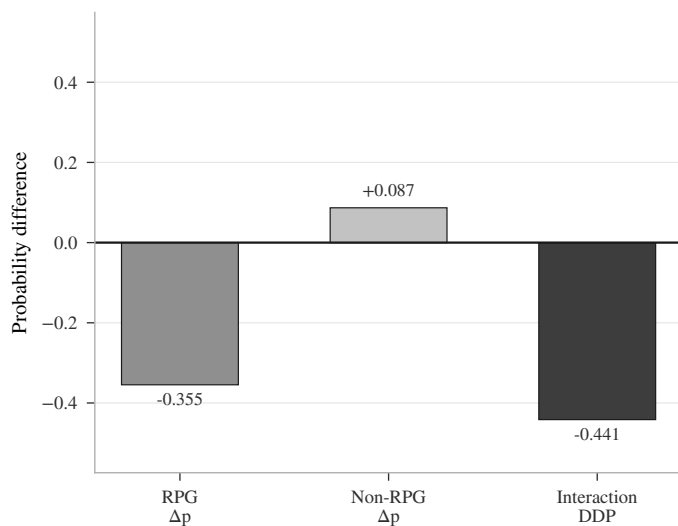


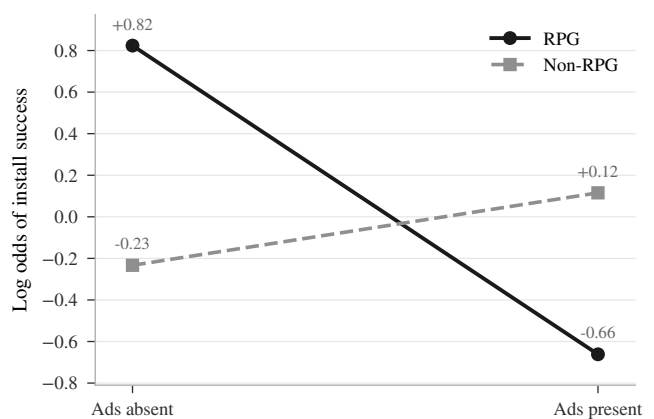
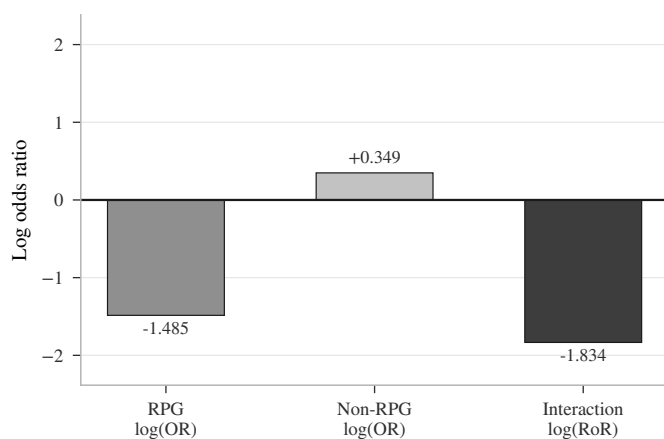
Figure 1. Additive interaction. The left panel decomposes the two within-stratum probability differences and their difference-in-differences; the right panel shows the corresponding conditional success probabilities.

4 Multiplicative interaction

$$RoR = OR_{RPG} / OR_{Non-RPG}$$

$$= 0.2264 / 1.4170 = \mathbf{0.1598}$$

95% CI: [0.1560, 0.1637]



Cross-market replication

Install success | United States and India

Country-specific calculations

United States

Stratum	Ads+ Success	Ads+ No	Ads- Success	Ads- No
RPG	629	724	1,437	388
Non-RPG	12,262	11,554	6,854	8,502

$$\Delta p_{\text{RPG}} = 0.4649 - 0.7874 = -0.3225$$

$$\Delta p_{\text{Non-RPG}} = 0.5149 - 0.4463 = +0.0685$$

$$\text{DDP} = -0.3225 - (+0.0685) = \mathbf{-0.3910}$$

$$\text{RoR} = 0.2346/1.3165 = \mathbf{0.1782}$$

Joint: --

India

Stratum	Ads+ Success	Ads+ No	Ads- Success	Ads- No
RPG	228	1,489	712	749
Non-RPG	15,746	12,453	4,496	6,477

$$\Delta p_{\text{RPG}} = 0.1328 - 0.4873 = -0.3545$$

$$\Delta p_{\text{Non-RPG}} = 0.5584 - 0.4097 = +0.1487$$

$$\text{DDP} = -0.3545 - (+0.1487) = \mathbf{-0.5032}$$

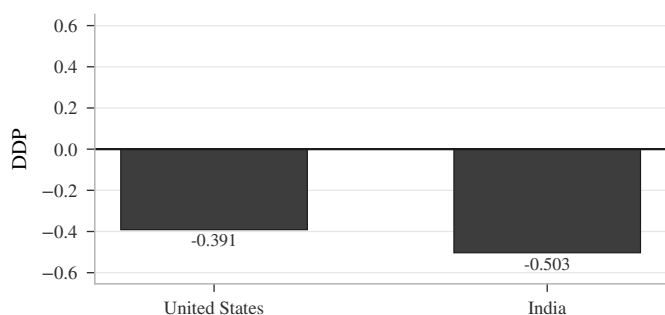
$$\text{RoR} = 0.1611/1.8216 = \mathbf{0.0884}$$

Joint: --

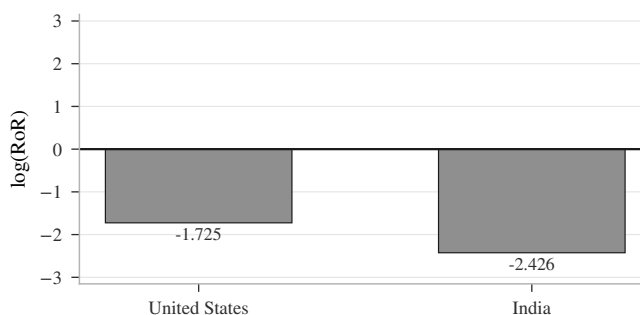
Comparison of interaction estimates

Context	RPG Δp	Non-RPG Δp	DDP	RPG OR	Non-RPG OR	RoR	Joint
United States	-0.323	+0.069	-0.391	0.235	1.316	0.178	--
India	-0.355	+0.149	-0.503	0.161	1.822	0.088	--

Additive interaction



Multiplicative interaction



Temporal replication

Install success | Three United States monthly contexts

Selected monthly contexts

Jan 2025

Stratum	Ads+ Success	Ads+ No	Ads- Success	Ads- No
RPG	51	45	96	35
Non-RPG	897	863	469	569

$$\Delta p_{\text{RPG}} = 0.5312 - 0.7328 = -0.2016$$

$$\Delta p_{\text{Non-RPG}} = 0.5097 - 0.4518 = +0.0578$$

$$\text{DDP} = -0.2016 - (+0.0578) = \mathbf{-0.2594}$$

$$\text{RoR} = 0.4132/1.2610 = \mathbf{0.3277}$$

Joint: --

Aug 2025

Stratum	Ads+ Success	Ads+ No	Ads- Success	Ads- No
RPG	41	53	96	37
Non-RPG	885	829	491	593

$$\Delta p_{\text{RPG}} = 0.4362 - 0.7218 = -0.2856$$

$$\Delta p_{\text{Non-RPG}} = 0.5163 - 0.4530 = +0.0634$$

$$\text{DDP} = -0.2856 - (+0.0634) = \mathbf{-0.3490}$$

$$\text{RoR} = 0.2982/1.2893 = \mathbf{0.2312}$$

Joint: --

Feb 2026

Stratum	Ads+ Success	Ads+ No	Ads- Success	Ads- No
RPG	44	69	91	23
Non-RPG	897	830	481	590

$$\Delta p_{\text{RPG}} = 0.3894 - 0.7982 = -0.4089$$

$$\Delta p_{\text{Non-RPG}} = 0.5194 - 0.4491 = +0.0703$$

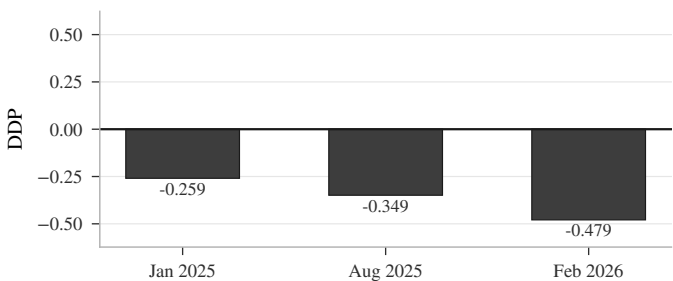
$$\text{DDP} = -0.4089 - (+0.0703) = \mathbf{-0.4791}$$

$$\text{RoR} = 0.1612/1.3256 = \mathbf{0.1216}$$

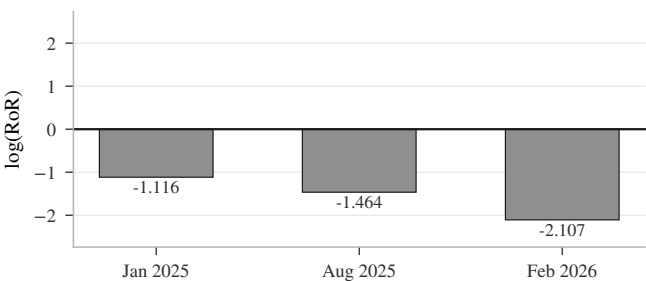
Joint: --

Temporal comparison

Additive interaction



Multiplicative interaction



Install-success synthesis

Pooled, national, and temporal contexts

Interaction estimates

Context	RPG Δp	Non-RPG Δp	DDP	RPG OR	Non-RPG OR	RoR	Joint
Pooled	-0.355	+0.087	-0.441	0.226	1.417	0.160	--
United States	-0.323	+0.069	-0.391	0.235	1.316	0.178	--
India	-0.355	+0.149	-0.503	0.161	1.822	0.088	--
US Jan 2025	-0.202	+0.058	-0.259	0.413	1.261	0.328	--
US Aug 2025	-0.286	+0.063	-0.349	0.298	1.289	0.231	--
US Feb 2026	-0.409	+0.070	-0.479	0.161	1.326	0.122	--

Interaction magnitude across contexts

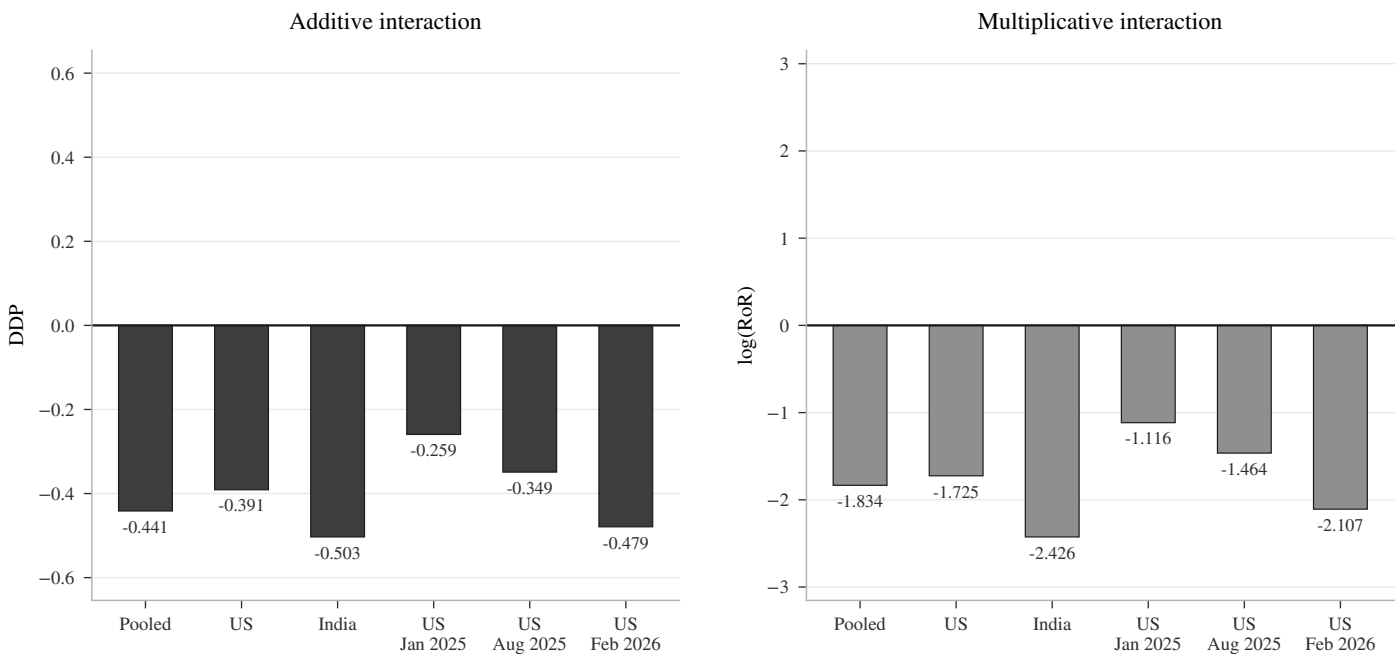


Figure 2. Install-success interaction estimates across the pooled analysis, two national markets, and three United States monthly contexts. Both panels use a zero null: DDP on the additive scale and log(RoR) on the multiplicative scale.

The worked contexts are presented to demonstrate how the same interaction calculation is propagated across analytical levels. The table retains RoR on its original ratio scale for reporting, whereas the figure uses log(RoR) so that the multiplicative null is centered at zero.

Parallel outcome analysis

Outcome 2 — Rating success | Pooled analysis

1 Construct the two exposure-by-outcome tables

RPG				Non-RPG			
	Ads = 1	Ads = 0	Total		Ads = 1	Ads = 0	Total
Success	27,224	48,847	76,071	Success	498,023	326,971	824,994
Not success	32,484	24,921	57,405	Not success	548,963	271,267	820,230
Total	59,708	73,768	133,476	Total	1,046,986	598,238	1,645,224

2 Estimate within-stratum associations

Stratum	P(success Ads = 1)	P(success Ads = 0)	Δp	OR
RPG	0.4560	0.6622	-0.2062	0.4276
Non-RPG	0.4757	0.5466	-0.0709	0.7527

3 Estimate interaction on both scales

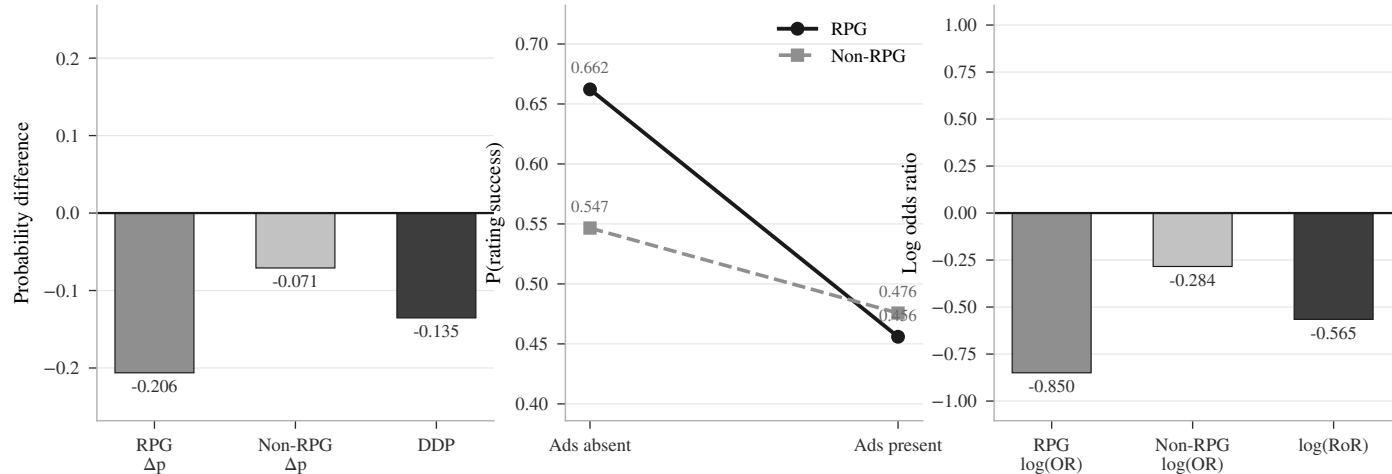
$DDP = -0.2062 - (-0.0709) = -0.1353$

$RoR = 0.4276/0.7527 = 0.5681$

95% CI: [-0.1408, -0.1298]

95% CI: [0.5551, 0.5814]

Joint classification: --



Rating-success replication

One national context and one monthly context

Context-specific calculations

United States

Stratum	Ads+ Success	Ads+ No	Ads- Success	Ads- No
RPG	643	710	1,260	565
Non-RPG	11,131	12,685	8,453	6,903

$$\Delta p_{\text{RPG}} = 0.4752 - 0.6904 = -0.2152$$

$$\Delta p_{\text{Non-RPG}} = 0.4674 - 0.5505 = -0.0831$$

$$\text{DDP} = -0.2152 - (-0.0831) = \mathbf{-0.1321}$$

$$\text{RoR} = 0.4061/0.7166 = \mathbf{0.5667}$$

Joint: --

United States — Aug 2025

Stratum	Ads+ Success	Ads+ No	Ads- Success	Ads- No
RPG	50	44	93	40
Non-RPG	786	928	598	486

$$\Delta p_{\text{RPG}} = 0.5319 - 0.6992 = -0.1673$$

$$\Delta p_{\text{Non-RPG}} = 0.4586 - 0.5517 = -0.0931$$

$$\text{DDP} = -0.1673 - (-0.0931) = \mathbf{-0.0742}$$

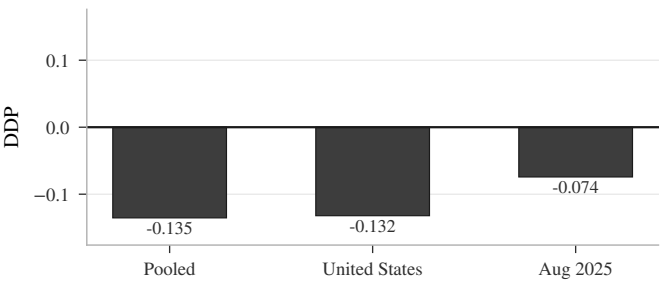
$$\text{RoR} = 0.4888/0.6884 = \mathbf{0.7100}$$

Joint: 00

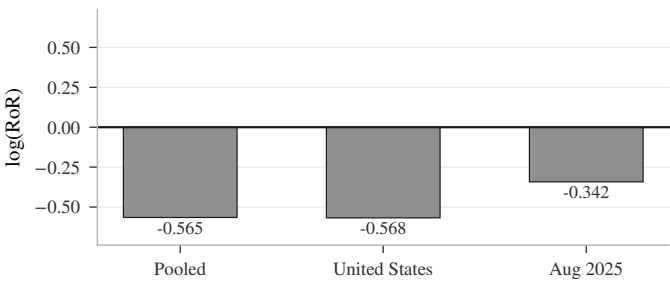
Rating-success summary

Context	RPG Δp	Non-RPG Δp	DDP	RPG OR	Non-RPG OR	RoR	Joint
Pooled	-0.206	-0.071	-0.135	0.428	0.753	0.568	--
United States	-0.215	-0.083	-0.132	0.406	0.717	0.567	--
US Aug 2025	-0.167	-0.093	-0.074	0.489	0.688	0.710	00

Additive interaction



Multiplicative interaction



Complete Ads × RPG interaction summary

Pooled, cross-market, and temporal evidence

Interaction estimates across analytical levels

Outcome	Analytical level	Mean / pooled DDP	95% CI / observed range	Mean / pooled RoR	Mean / pooled log(RoR)	++	--	00	Mixed	N
Install	Pooled	-0.441	[-0.447, -0.436]	0.160	-1.834	0	1	0	0	1
Install	Countries	-0.421	-0.525 to -0.313	0.163	-1.852	0	42	0	0	42
Install	Country × month	-0.422	-0.649 to -0.172	0.168	-1.869	0	588	0	0	588
Rating	Pooled	-0.135	[-0.141, -0.130]	0.568	-0.565	0	1	0	0	1
Rating	Countries	-0.127	-0.205 to -0.062	0.587	-0.539	0	42	0	0	42
Rating	Country × month	-0.127	-0.353 to +0.092	0.611	-0.544	0	291	292	5	588

Pooled rows report the interaction estimate and its 95% confidence interval. Country and country × month rows summarize the complete set of context-specific estimates; the interval column therefore shows the observed range rather than a confidence interval.

Cross-market classification

Outcome	Mean DDP	Mean RoR	++	--	00	Mixed	Countries
Install success	-0.421	0.163	0	42	0	0	42
Rating success	-0.127	0.587	0	42	0	0	42

Country × month classification

Outcome	Mean DDP	Mean RoR	++	--	00	Mixed	Contexts
Install success	-0.422	0.168	0	588	0	0	588
Rating success	-0.127	0.611	0	291	292	5	588

Joint classification combines additive and multiplicative inference: ++ indicates statistically positive interaction on both scales; -- indicates statistically negative interaction on both scales; 00 indicates null interaction on both scales; Mixed contains all remaining combinations. Multiplicative effects are reported as RoR in the tables and transformed to log(RoR) when a zero-centered visual scale is required.

Country-specific interaction estimates

Ads × RPG interaction | All national markets

Install and rating interaction estimates by country

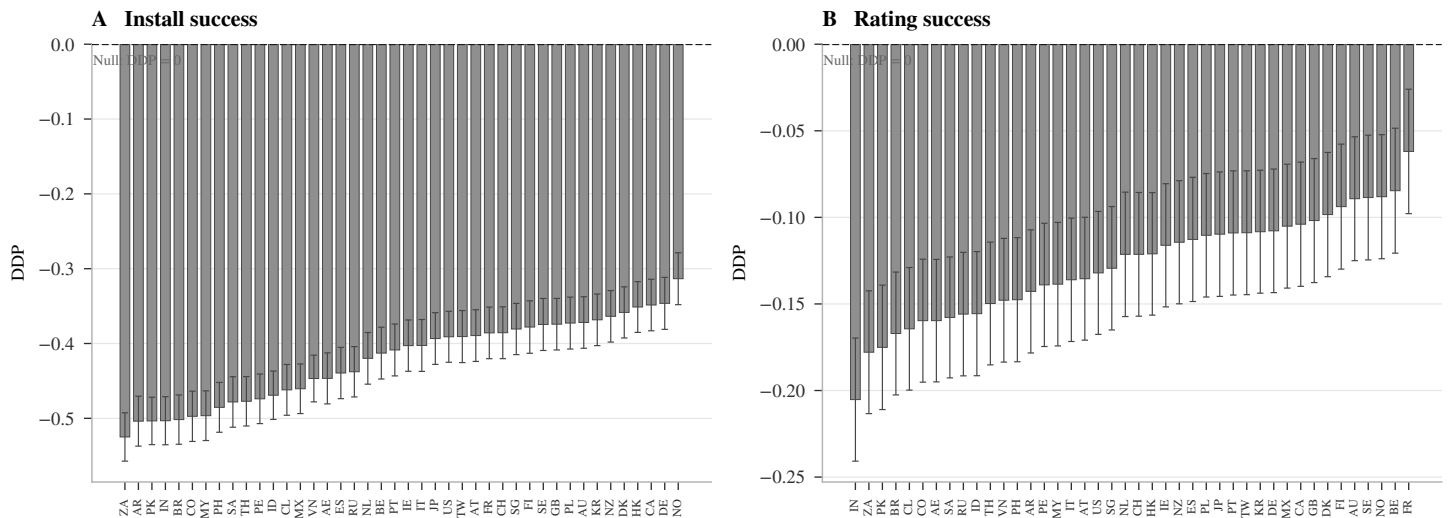
Country	Install DDP [95% CI]	Install RoR [95% CI]	Rating DDP [95% CI]	Rating RoR [95% CI]
AE	-0.447 [-0.481, -0.413]	0.151 [0.129, 0.177]	-0.160 [-0.195, -0.124]	0.509 [0.438, 0.592]
AR	-0.504 [-0.537, -0.470]	0.118 [0.101, 0.139]	-0.143 [-0.178, -0.107]	0.556 [0.479, 0.645]
AT	-0.389 [-0.424, -0.355]	0.187 [0.160, 0.219]	-0.135 [-0.171, -0.100]	0.552 [0.474, 0.643]
AU	-0.372 [-0.406, -0.337]	0.197 [0.168, 0.231]	-0.089 [-0.125, -0.053]	0.679 [0.584, 0.789]
BE	-0.413 [-0.447, -0.378]	0.176 [0.151, 0.206]	-0.085 [-0.121, -0.049]	0.700 [0.603, 0.813]
BR	-0.502 [-0.535, -0.469]	0.112 [0.095, 0.132]	-0.167 [-0.203, -0.132]	0.504 [0.434, 0.585]
CA	-0.349 [-0.383, -0.314]	0.212 [0.180, 0.248]	-0.104 [-0.140, -0.068]	0.639 [0.549, 0.743]
CH	-0.386 [-0.420, -0.351]	0.185 [0.158, 0.217]	-0.121 [-0.157, -0.086]	0.584 [0.501, 0.681]
CL	-0.462 [-0.496, -0.428]	0.146 [0.125, 0.170]	-0.164 [-0.200, -0.129]	0.507 [0.437, 0.589]
CO	-0.497 [-0.531, -0.464]	0.122 [0.104, 0.142]	-0.160 [-0.195, -0.124]	0.520 [0.448, 0.603]
DE	-0.346 [-0.381, -0.312]	0.223 [0.191, 0.261]	-0.108 [-0.143, -0.072]	0.624 [0.537, 0.727]
DK	-0.358 [-0.393, -0.324]	0.195 [0.165, 0.229]	-0.098 [-0.134, -0.063]	0.649 [0.557, 0.756]
ES	-0.439 [-0.474, -0.405]	0.160 [0.137, 0.187]	-0.113 [-0.149, -0.077]	0.626 [0.539, 0.727]
FI	-0.378 [-0.413, -0.343]	0.202 [0.173, 0.236]	-0.094 [-0.130, -0.058]	0.670 [0.576, 0.779]
FR	-0.386 [-0.420, -0.351]	0.187 [0.160, 0.219]	-0.062 [-0.098, -0.026]	0.762 [0.655, 0.885]
GB	-0.374 [-0.409, -0.340]	0.189 [0.161, 0.222]	-0.102 [-0.138, -0.066]	0.638 [0.548, 0.743]
HK	-0.351 [-0.385, -0.317]	0.203 [0.173, 0.239]	-0.121 [-0.156, -0.086]	0.586 [0.503, 0.682]
ID	-0.469 [-0.501, -0.437]	0.108 [0.091, 0.129]	-0.156 [-0.191, -0.120]	0.531 [0.458, 0.616]
IE	-0.403 [-0.437, -0.369]	0.169 [0.144, 0.198]	-0.116 [-0.152, -0.081]	0.596 [0.511, 0.694]
IN	-0.503 [-0.535, -0.471]	0.088 [0.074, 0.106]	-0.205 [-0.241, -0.170]	0.431 [0.370, 0.501]
IT	-0.403 [-0.437, -0.368]	0.185 [0.158, 0.216]	-0.136 [-0.172, -0.100]	0.562 [0.483, 0.653]
JP	-0.393 [-0.428, -0.359]	0.181 [0.155, 0.212]	-0.110 [-0.146, -0.074]	0.624 [0.536, 0.726]
KR	-0.368 [-0.403, -0.334]	0.200 [0.171, 0.235]	-0.108 [-0.144, -0.073]	0.619 [0.531, 0.720]
MX	-0.460 [-0.494, -0.427]	0.134 [0.114, 0.157]	-0.105 [-0.141, -0.069]	0.651 [0.562, 0.755]
MY	-0.496 [-0.530, -0.463]	0.119 [0.101, 0.140]	-0.139 [-0.174, -0.103]	0.568 [0.490, 0.659]
NL	-0.420 [-0.454, -0.385]	0.174 [0.149, 0.204]	-0.121 [-0.157, -0.085]	0.602 [0.518, 0.700]
NO	-0.313 [-0.348, -0.279]	0.244 [0.208, 0.287]	-0.088 [-0.124, -0.052]	0.671 [0.575, 0.782]
NZ	-0.364 [-0.398, -0.329]	0.202 [0.172, 0.236]	-0.114 [-0.150, -0.079]	0.604 [0.519, 0.703]
PE	-0.474 [-0.507, -0.441]	0.123 [0.104, 0.145]	-0.139 [-0.175, -0.103]	0.565 [0.487, 0.656]
PH	-0.485 [-0.519, -0.452]	0.115 [0.097, 0.136]	-0.148 [-0.183, -0.112]	0.548 [0.472, 0.636]
PK	-0.503 [-0.535, -0.472]	0.075 [0.062, 0.090]	-0.175 [-0.211, -0.139]	0.486 [0.418, 0.565]
PL	-0.373 [-0.407, -0.338]	0.209 [0.179, 0.244]	-0.110 [-0.146, -0.075]	0.627 [0.540, 0.728]
PT	-0.409 [-0.443, -0.374]	0.184 [0.158, 0.215]	-0.109 [-0.145, -0.073]	0.633 [0.545, 0.735]
RU	-0.438 [-0.471, -0.404]	0.154 [0.131, 0.181]	-0.156 [-0.192, -0.120]	0.527 [0.454, 0.612]
SA	-0.478 [-0.512, -0.444]	0.132 [0.113, 0.155]	-0.158 [-0.193, -0.123]	0.505 [0.434, 0.588]
SE	-0.375 [-0.409, -0.340]	0.201 [0.171, 0.235]	-0.089 [-0.125, -0.053]	0.684 [0.588, 0.795]
SG	-0.381 [-0.415, -0.346]	0.181 [0.154, 0.213]	-0.129 [-0.165, -0.094]	0.567 [0.487, 0.661]
TH	-0.477 [-0.510, -0.444]	0.126 [0.107, 0.148]	-0.150 [-0.185, -0.114]	0.540 [0.466, 0.627]
TW	-0.391 [-0.426, -0.356]	0.197 [0.169, 0.230]	-0.109 [-0.145, -0.073]	0.632 [0.544, 0.734]
US	-0.391 [-0.425, -0.357]	0.178 [0.152, 0.209]	-0.132 [-0.168, -0.097]	0.567 [0.487, 0.659]
VN	-0.447 [-0.478, -0.416]	0.104 [0.086, 0.125]	-0.148 [-0.184, -0.112]	0.547 [0.472, 0.635]
ZA	-0.525 [-0.557, -0.493]	0.092 [0.077, 0.109]	-0.178 [-0.213, -0.142]	0.482 [0.415, 0.560]

Each row reports the country-specific Ads × RPG interaction estimate and its 95% confidence interval. DDP is the additive interaction (null = 0); RoR is the multiplicative interaction (null = 1). The table reports the complete national-market estimates rather than cross-country means.

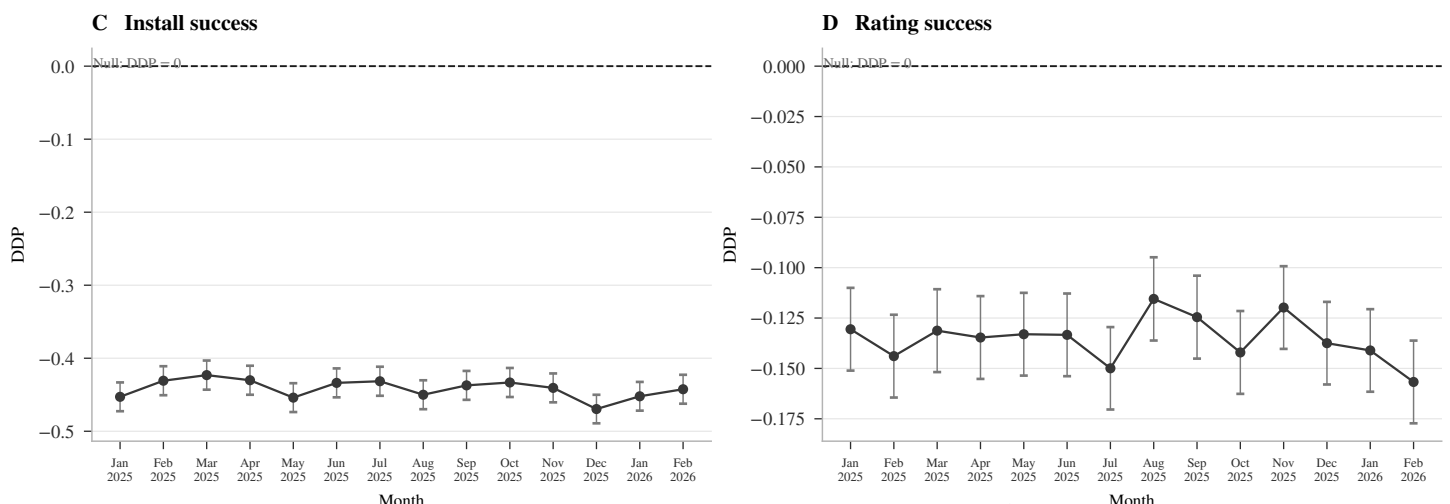
Complete cross-market and temporal synthesis

Ads × RPG interaction | Install success and rating success

Cross-market interaction — all national markets



Temporal interaction — all 14 pooled months



Reading the figure

DDP = 0 is the additive no-interaction reference. Values below zero indicate that the Ads-associated probability difference is more negative for RPG than for non-RPG games; values above zero indicate the reverse. The corresponding multiplicative evidence is retained in the complete numerical synthesis on the preceding page using RoR and log(RoR).